A National Dataset of Downtown Business District Boundaries for Pre-Automobile U.S. Cities and Towns

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Abstract

The downtown business district is a fundamental unit of analysis in urban revitalization research and local economic development, yet no nationally consistent, publicly available dataset of downtown boundaries exists for small and mid-sized U.S. cities and towns. This paper introduces an open data product containing approximately 6,500 downtown polygons for pre-automobile-era places spanning 47 contiguous states. The dataset is constructed from a fully reproducible, open-source R workflow: a national universe of historically significant places is assembled using historical census records, contemporary population data, and metropolitan classification criteria; points-of-interest data from the Overture Maps Foundation are then collected within each place boundary; and a kernel density estimation algorithm identifies the highest-concentration commercial cluster as the downtown core. The resulting polygons are deposited at Harvard Dataverse under a CC0 1.0 license, with all R code available on GitHub. The dataset is well-suited for research on downtown revitalization, Main Street commercial activity, and the economic geography of small and mid-sized American cities.

1. Introduction

The downtown business district is the historic commercial core, or "heart," of the American city (Murphy and Vance Jr, 1954), concentrating offices, retail, and the highest pedestrian flows and land values in the urban system (Van Leuven, 2026a). For most cities and towns—particularly those that grew before the automobile era—it was where residents shopped, banked, interacted with local government, and gathered socially. Research on downtown revitalization has grown substantially in recent decades, producing a body of work that examines what conditions predict recovery (Walzer and Kline, 2013), which policies are effective (Van Leuven, 2022a, 2024), and how downtown vitality relates to broader urban economic and demographic trends (Faulk, 2006).

However, this research has a persistent data problem: there is no nationally consistent, publicly available dataset of downtown boundaries for small and mid-sized cities. Prior efforts to stan-

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dardize CBD delimitation have remained limited in scale. Classic methods constructed detailed, block-level indices of central business intensity for individual cities (Murphy and Vance Jr, 1954), while more recent computational work applies kernel density estimation to points-of-interest data (Yu et al., 2015), topology-based methods to geospatial big data (Ren et al., 2024), or open-data workflows to produce commercial boundary datasets for large metropolitan areas (Jeong et al., 2024). None of these approaches has been extended to thousands of smaller pre-automobile towns. In practice, studies typically define downtown boundaries by hand for individual places (see Main Street America, 2026), rely on locally administered district boundaries such as business improvement districts or Main Street program areas, or use census geographies—tracts, block groups, or ZIP codes—that were not designed to capture commercial cores. None of these approaches transfers consistently across thousands of places, and none is reproducible in a way that allows other researchers to extend or replicate the work.

This paper introduces a dataset that addresses this gap directly. The dataset contains downtown polygons for approximately 6,500 pre-automobile-era cities and towns across 47 contiguous states, deposited at Harvard Dataverse under a CC0 1.0 license (Van Leuven, 2026b). The workflow that generates them is fully reproducible and draws on open data sources available for every city in the country: historical population records, U.S. Census place boundaries, and points-of-interest data from the Overture Maps Foundation (Overture Maps Foundation, 2024). All R code is available on GitHub.

The present work extends an earlier delineation method proposed in Van Leuven (2022b), which relied on parcel and business license records that are not consistently available at national scale. By substituting universally available points-of-interest data, the revised method can be applied to any incorporated place with a sufficient commercial presence, making national coverage possible.

2. The Pre-Automobile Town Universe

Not every U.S. city or town is a suitable candidate for downtown delineation. Very large cities have commercial cores too complex for a single-polygon approach; very small or recently settled places may lack a recognizable commercial district altogether. The dataset targets a specific and theoretically motivated population: places that developed commercial cores before the automobile era.

The rationale for this focus is straightforward. A place that reached at least 750 residents between 1900 and 1940 almost certainly organized its commerce around a walkable main street or town square. Field studies of small-town commercial morphology confirm that early CBDs typically formed at pre-railroad crossroads or courthouse squares and continue to host the majority of retail and professional establishments in these communities even where some activity has migrated to outlying highway corridors (Lamb, 1985). That physical legacy—the traditional street grid, the historic building stock, the concentration of storefronts—persists in many of these places today and defines the territory this dataset is meant to capture. Places that did not reach this threshold before the automobile era tend to have commercial development organized around arterial roads or highway interchanges rather than a traditional downtown.

Four criteria define membership in the pre-automobile universe, closely following Van Leuven (2022b):

1. *Historical population.* The place had a population of at least 750 residents in at least one decennial census year from 1900 to 1940, based on historical population series compiled from Wikipedia and hosted in the Historical-Populations GitHub project (CreatingData, 2023).

2. *Contemporary population.* The place had a 2020 Census population between 500 and 150,000. This excludes ghost towns that have largely depopulated, as well as large cities for which a different delineation approach is more appropriate.

3. *Metropolitan classification.* The place is not located in a central county of a large metropolitan area, based on Rural-Urban Continuum Codes from the USDA Economic Research Service (2024) combined with OMB metropolitan area delineation files. This filter excludes core suburban municipalities, while retaining places in outlying counties of large metro areas, which often contain genuinely independent historic commercial districts.[1]

4. *Place type.* Census Designated Places (CDPs) are excluded by default, as they typically lack incorporated boundaries and the formal administrative structure that tends to accompany historic commercial cores. An exception is made for New England and several township-heavy states—Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont, New York, New Jersey, Pennsylvania, Michigan, Minnesota, and Wisconsin—where CDPs frequently represent genuine historic town centers.

Applied nationally, these criteria yield approximately 6,500 places distributed across 47 states. Figure 1 maps the universe. Coverage is densest in the Midwest, Northeast, and South—regions most fully settled during the pre-automobile era. The universe is assembled in R using place-level

[1]Newark, Ohio illustrates the rationale for this exception. Licking County, which contains Newark, has a RUCC code of 1 (counties in metro areas of one million or more), which would ordinarily trigger exclusion. However, Licking County is classified as an *outlying* rather than *central* county in the Columbus MSA, so it passes the filter. Newark has a functioning pre-automobile courthouse square downtown that is entirely distinct in character from Columbus's suburbs—retaining it while excluding Columbus and its automobile-era municipalities is precisely the intended behavior of the combined RUCC/OMB criterion.

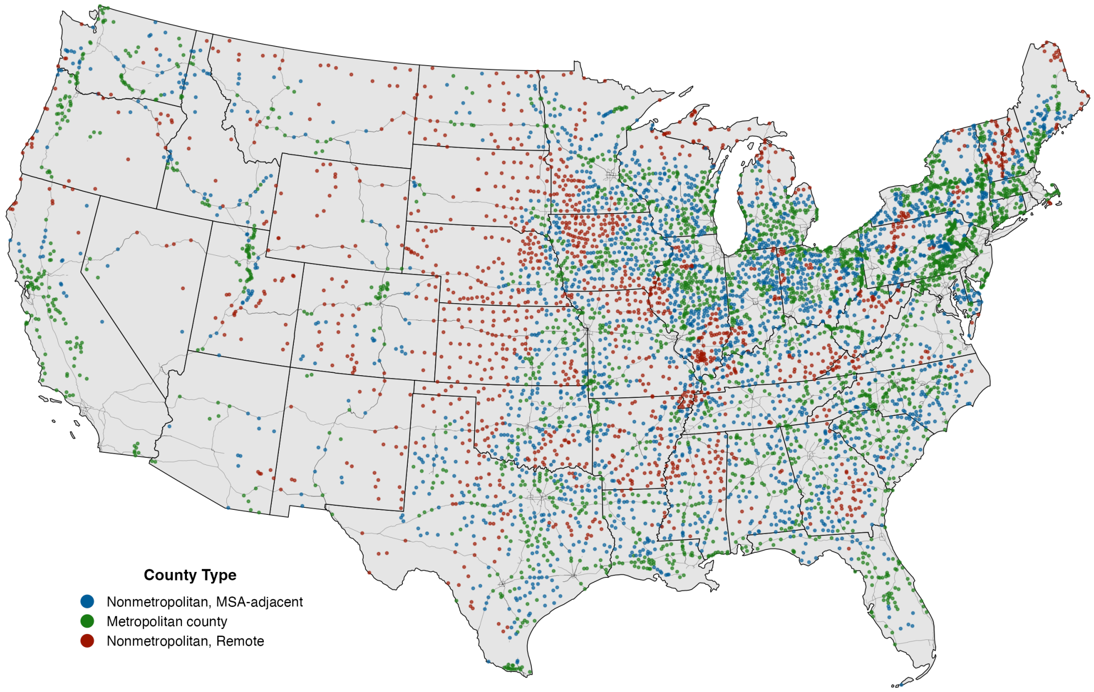


Figure 1: *The pre-automobile town universe. Each point represents one of the approximately 6,500 places included in the dataset, distributed across 47 contiguous states.*

population data and boundaries from `tidycensus` (Walker and Herman, 2023) and `tigris` (Walker, 2023), with county-level RUCC and OMB classifications joined to each place via its host county.

3. Delineation Method

For each place in the pre-automobile universe, a downtown polygon is generated in three steps: POI data collection, kernel density estimation, and polygon construction. The theoretical and methodological case for using kernel density estimation to approximate downtown business district boundaries is developed in Van Leuven (2022b); the contribution of the present paper is to operationalize that strategy at national scale using Overture Maps POI data, which are openly available for every city in the country and eliminate the dependence on locally held establishment records that limited the earlier approach.

POI data collection. Points-of-interest data are drawn from the Overture Maps Foundation's Places dataset (Overture Maps Foundation, 2024), a large open database of business and amenity locations assembled from multiple commercial and open sources. For each state, POIs are extracted from a vector tile archive using the `pmtiles` command-line tool and then spatially intersected with pre-automobile place boundaries to assign each POI to a town.

Some POI categories are poor indicators of downtown commercial activity because they represent institutional rather than retail or service functions. Large institutional facilities—college campuses, hospitals, and government office complexes—tend to generate dense POI clusters that can overwhelm the commercial signal in the density estimation step, particularly in small cities where a single institution is among the largest landholders. These POIs are flagged using category metadata and excluded from density calculations by default.

Kernel density estimation. Following CBD research that treats centrality as a smoothed density surface over point-of-interest data (Yu et al., 2015), a kernel density estimate is computed on a hexagonal grid for each place using the `sfhotspot` package (Ashby, 2025), which wraps the `SpatialKDE` backend. Two key parameters—cell size and bandwidth—are set automatically. Cell size is chosen so that 50 cells span the shorter dimension of the place's bounding box, rounded to the nearest 100 metres; for a typical small city, this yields cells of roughly 25–50 metres on a side. Bandwidth is selected via Silverman's rule of thumb (Silverman, 1986), implemented through R's `bandwidth.nrd` function, which derives the smoothing radius from the standard deviation and interquartile range of the input point coordinates; bandwidths of roughly 150–250 metres are common across the universe of places. The quartic kernel function is used throughout. The hexagonal grid provides even spatial coverage without the edge artifacts common in square-grid approaches. Hexagonal cells in the top 25 percent of the min-max scaled KDE distribution are retained as candidate downtown cells.

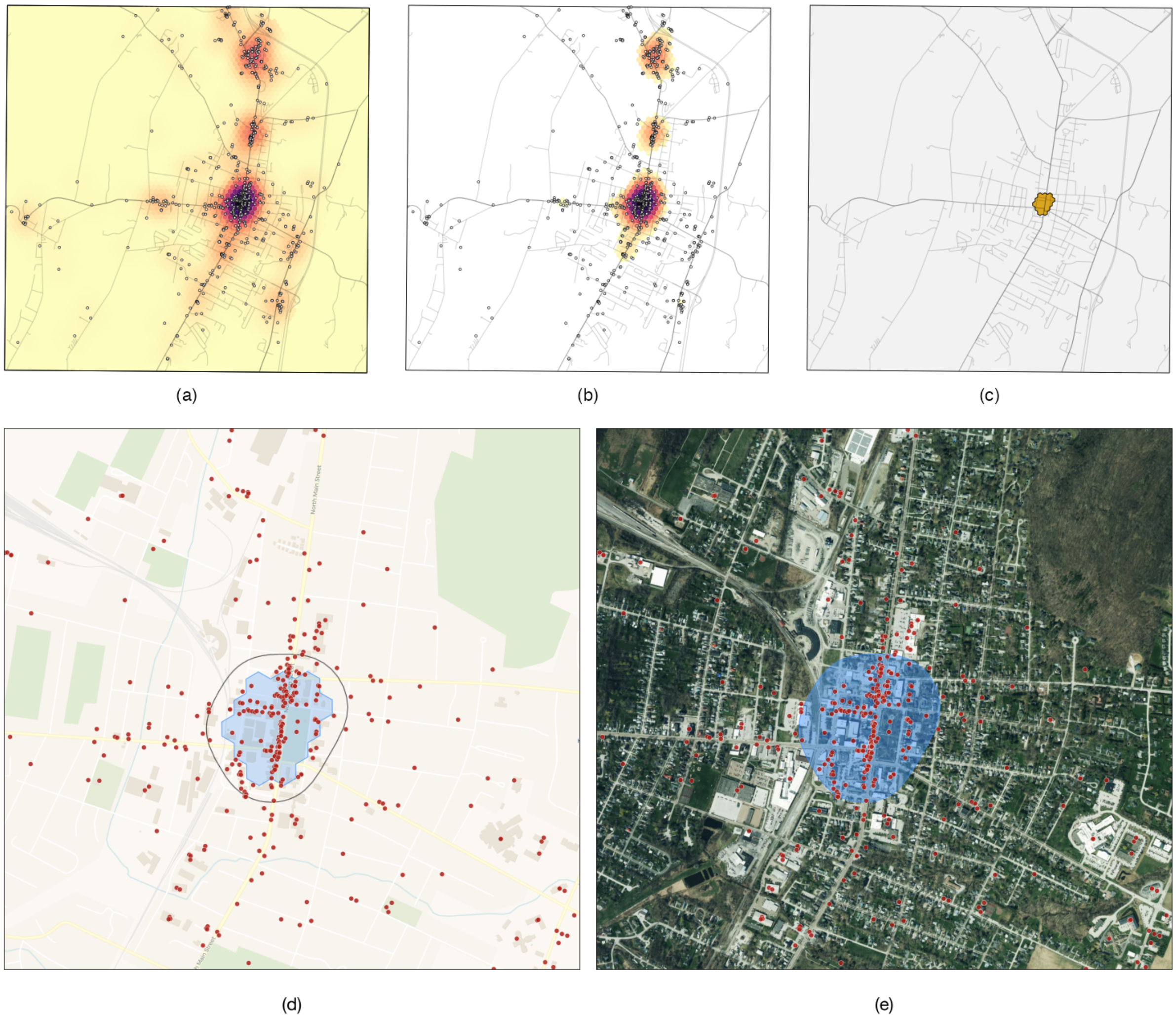
Polygon construction. Retained cells are dissolved into contiguous clusters, and each cluster is scored by multiplying its mean KDE intensity by its cell count. This composite score rewards clusters that are both dense and spatially extensive—the two properties most characteristic of a traditional downtown commercial district. The highest-scoring cluster is selected as the downtown core. The cluster boundary is then buffered slightly and smoothed to remove jagged edges, producing the final polygon.

Figure 2 illustrates the full delineation sequence for St. Albans, Vermont. Each panel corresponds to a step in the algorithm, from initial density estimation through to the final polygon output.

4. Dataset Description and Illustrative Application

The final dataset contains one polygon per city. The primary output file, `downtowns_us.geojson`, contains all approximately 6,500 features in a single national file; per-state GeoJSON files are also provided.[2] The dataset is accompanied by two companion files. `national_pre_auto_places.csv` contains the full pre-automobile universe with place identifiers, population figures, and metropoli-

[2] Table A1 in Appendix A describes the fields included with each feature.



Panel	Description
a	Kernel density estimate computed on a hexagonal grid across the full place boundary, with Overture Maps points-of-interest overlaid; each hexagon is colored by its KDE value.
b	Hexagons whose KDE values exceed one standard deviation above the mean.
c	Top-25 percent of hexagons by min-max scaled KDE value, dissolved into contiguous candidate clusters; the highest-scoring cluster is selected as the downtown core.
d	Selected cluster and POI locations on a street basemap at closer range, with the final smoothed polygon boundary shown in black.
e	Final polygon on satellite imagery, confirming that the delineated boundary closely follows the walkable historic commercial core.

Figure 2: *Illustration of the downtown delineation method for Saint Albans, Vermont*

tan classification variables. `pre_auto_poi_us.csv` contains the full national POI dataset used in density estimation, including each point's town assignment, category, institutional flag, and coordinates. Following the open data products framework (Arribas-Bel et al., 2021), all files are deposited at Harvard Dataverse under a CC0 1.0 Universal license (Van Leuven, 2026b), placing no restrictions on reuse. The R workflow consists of four scripts that, respectively, build the national universe, collect POI data for a single state, run the delineation algorithm for a single state, and batch-process all 48 contiguous states. All R code is available on GitHub.[3]

The delineation output for St. Albans, Vermont shown in Figure 2 is illustrative of the dataset more broadly: the algorithm reliably identifies a compact, walkable commercial core from points-of-interest distributed across a much larger place boundary.[4] Across the full universe of places, the dataset exhibits meaningful variation in polygon size and shape that reflects real differences in commercial geography. Larger and more economically active cities tend to produce larger polygons with higher KDE intensity scores. Cities with more dispersed commercial activity—those whose historic main street was interrupted by highway construction or urban renewal, for example—sometimes produce smaller or more irregular polygons.

5. Conclusion

This paper introduces a nationally consistent, publicly available dataset of downtown business district boundaries for approximately 6,500 pre-automobile-era cities and towns across the contiguous United States. The dataset fills a long-standing gap in the empirical infrastructure available to researchers studying downtown revitalization, Main Street commercial activity, and the economic geography of small and mid-sized American places. By combining historical census records, contemporary place boundaries, and openly available points-of-interest data within a fully reproducible workflow, the method produces polygon boundaries that are consistent in their construction, grounded in the commercial geography of each place, and extensible as new data become available. That said, like any data product built from open sources and applied at national scale, this one carries limitations that users should understand before applying it in research or practice.

5.1 Data Limitations

Three limitations of this dataset are worth noting. First, the quality and completeness of Overture Maps POI data varies across places. Rural and lower-income communities tend to be un-

[3]GitHub repository URL: `https://github.com/andrewvanleuven/downtown_polygons`.

[4]Additional examples drawn from geographically and economically diverse settings—Morton, WA and Mullins, SC—are provided in Figure B1 in Appendix B.

derrepresented in commercial POI databases relative to more densely populated or economically active areas. Uniform underrepresentation across a city is unlikely to distort the delineation result, since the density estimation is sensitive to relative, rather than absolute, POI counts. More consequential would be *systematic* underrepresentation of downtown businesses specifically, or systematic overrepresentation of large-format retail outside the historic core. Users working in rural or lower-income contexts should visually inspect polygon results before relying on them analytically.

Second, the dataset represents a snapshot in time. The POI data and Census boundaries used in the workflow reflect conditions around 2024, and the commercial landscape of any given city will shift as businesses open and close. Because the workflow is fully open source and Overture Maps releases periodic updates to its POI data, the delineation can, in principle, be rerun as new data become available. In practice, however, the physical location and spatial extent of the downtown district in a pre-automobile city or town is unlikely to change substantially over time; the historic street grid and building stock that define these cores are largely fixed features of the urban landscape.

A third limitation concerns the scope of the universe itself. By design, the dataset excludes places with 2020 populations above 150,000. In large cities, postwar redevelopment transformed CBDs into internalized office and retail complexes—consolidated superblocks, corporate plazas, and enclosed malls—that differ substantially in physical form from the walkable, street-front commercial cores targeted here (Beauregard, 1986). Delineating downtown districts in such contexts is a categorically different methodological problem requiring different data and approaches. It is also a problem that has received considerably more scholarly and administrative attention, making a nationally consistent open dataset less urgently needed for larger places than for the small and mid-sized cities this dataset is designed to serve.

5.2 Future Directions & Applications

The most immediate application of this dataset is as a spatial filter for other data sources. A researcher studying business activity can intersect the polygon layer with establishment-level records to extract downtown businesses across the full pre-automobile universe. Similar operations are possible with parcel records, building permit data, Opportunity Zone designations, or pedestrian count data. The polygons can also serve as units of analysis for change-over-time studies by intersecting them with multi-year data sources such as the Longitudinal Employer-Household Dynamics dataset or American Community Survey tract estimates.

Beyond these foundational uses, the dataset opens several directions for substantive research on downtown revitalization and small-city economic geography. One natural application is the construction of longitudinal indicators of downtown commercial health. By intersecting the polygons with point-level microdata—such as SafeGraph foot traffic records or CoreLogic property data—researchers can track changes in commercial activity and real estate conditions within downtown boundaries over time. These indicators could support both descriptive accounts of downtown trajectories and causal analyses of policies designed to stimulate revitalization, such as historic tax credits, Main Street program participation, or Opportunity Zone designations. The polygons can also be combined with parcel-level data, building footprint datasets, or street network data to measure physical attributes of downtown cores—building age, lot coverage, street connectivity, and historic district designation—across thousands of places simultaneously, enabling systematic tests of hypotheses about urban form and commercial outcomes at a scale that has not previously been feasible.

The dataset also supports research on the demographic and socioeconomic context of downtown districts. Intersecting the polygons with Census tract or block group data allows researchers to characterize residential populations living in and around downtown areas, opening questions about gentrification and the distributional consequences of downtown investment in small and mid-sized cities. Transportation and accessibility researchers can similarly intersect the polygons with pedestrian infrastructure, transit access, or mobility data to assess how well contemporary networks serve these historically walkable cores. On the methodological side, the workflow could be extended temporally using historical POI sources—such as digitized city directories or Sanborn fire insurance maps—to reconstruct downtown boundaries at earlier points in time, or adapted geographically for pre-automobile towns in other countries where comparable open POI data are available.

The dataset also has direct applications beyond academic research. Economic development practitioners, Main Street program administrators, and municipal planners frequently lack consistent boundary definitions for the districts they manage, and a nationally consistent polygon layer provides a common spatial framework for benchmarking downtown performance and targeting interventions. More broadly, this dataset represents one piece of a larger empirical infrastructure that the field of downtown revitalization research has long lacked. By making both the data and the workflow openly available, this paper aims to lower the barriers to rigorous, reproducible, and comparative research on the places that sit at the heart of small-town American economic life.

Data Availability Statement

The downtown polygon dataset introduced in this paper, along with companion files containing the full pre-automobile place universe and the national points-of-interest data used in delineation, are publicly available at Harvard Dataverse (https://doi.org/10.7910/DVN/RFXZ3F) under a CC0 1.0 Universal license. All R code used to construct the dataset is available at https://github.com/andrewvanleuven/downtown_polygons.

References

Arribas-Bel, D., Green, M., Rowe, F., and Singleton, A. (2021). Open data products: A framework for creating valuable analysis-ready data. *Journal of Geographical Systems*, 23(4):497–514.

Ashby, M. (2025). `sfhotspot`: Hot-spot analysis with spatial features. R package version 1.0.0.

Beauregard, R. A. (1986). Urban form and the redevelopment of central business districts. *Journal of Architectural and Planning Research*, pages 183–198.

CreatingData (2023). Historical-populations: Per-state historical population series for U.S. cities and towns.

Faulk, D. (2006). The process and practice of downtown revitalization. *Review of policy research*, 23(2):625–645.

Jeong, B., Allen, J., and Chapple, K. (2024). A new commercial boundary dataset for metropolitan areas in the usa and canada, built from open data. *Scientific Data*, 11(1):422.

Lamb, R. F. (1985). The morphology and vitality of business districts in upstate new york villages. *The Professional Geographer*, 37(2):162–172.

Main Street America (2026). Building opportunities on main street (booms) tracker. https://booms-tracker-nthp.hub.arcgis.com/. An ArcGIS Hub-based tool for gathering, storing, and showcasing information about buildings and lots in local Main Street districts. Latest update: Spring 2026.

Murphy, R. E. and Vance Jr, J. E. (1954). Delimiting the cbd. *Economic Geography*, 30(3):189–222.

Overture Maps Foundation (2024). Overture maps places dataset.

Ren, Z., Seipel, S., and Jiang, B. (2024). A topology-based approach to identifying urban centers in america using multi-source geospatial big data. *Computers, Environment and Urban Systems*, 107:102045.

Silverman, B. W. (1986). *Density Estimation for Statistics and Data Analysis*. Chapman and Hall, London.

USDA Economic Research Service (2024). Rural-urban continuum codes. Accessed: 2026-03-20.

Van Leuven, A. J. (2022a). The impact of main street revitalization on the economic vitality of small-town business districts. *Economic Development Quarterly*, 36(3):193–207.

Van Leuven, A. J. (2022b). A method for defining downtown business district boundaries in pre-automobile towns and cities. *Cityscape*, 24(1):369–382.

Van Leuven, A. J. (2024). Leveraging main street as a real estate amenity: Downtown revitalization and residential property values. *Journal of Planning Education and Research*, 44(3):1488–1502.

Van Leuven, A. J. (2026a). Central business district. In *The Encyclopedia of Human Geography*, pages 105–110. Springer.

Van Leuven, A. J. (2026b). Downtown Business District Polygons for Pre-Automobile-Era U.S. Cities and Towns.

Walker, K. (2023). `tigris`: Load census TIGER/Line shapefiles. R package version 2.0.

Walker, K. and Herman, M. (2023). `tidycensus`: Load US census boundary and attribute data as `tidyverse` and `sf`-ready data frames. R package version 1.6.

Walzer, N. and Kline, S. (2013). An evaluation of approaches to downtown economic revitalization. In *Downtowns*, pages 249–274. Routledge.

Yu, W., Ai, T., and Shao, S. (2015). The analysis and delimitation of central business district using network kernel density estimation. *Journal of Transport Geography*, 45:32–47.

Appendices

Appendix A: Data Description

Table A1: *Fields in the primary output file (`downtowns_us.geojson`). Example values are for St. Albans, Vermont.*

Field	Type	Description	Example
Place identifiers			
`city_fips`	Character	7-digit Census place FIPS code (2-digit state + 5-digit place)	`5061675`
`city_name`	Character	Place name with Census type suffixes removed	`St. Albans`
`state`	Character	Two-letter postal state abbreviation	`VT`
`pop_20`	Integer	2020 Census total population of the incorporating place	`6,877`
Downtown delineation			
`n_poi`	Integer	Overture Maps POIs used in kernel density estimation (excludes college campuses and hospital complexes)	`369`
`kde_intensity`	Numeric	Mean KDE z-score across the hexagonal cells in the selected downtown cluster; higher values indicate a denser, more concentrated commercial core	`6.05`
`downtown_score`	Numeric	`kde_intensity` $\times$ cell count; the composite metric used to select the primary cluster when multiple candidates exist	`193.7`
Host county context			
`cty_fips`	Character	5-digit Census county FIPS code of the county containing the place	`50011`
`cty_type`	Character	OMB metropolitan area classification of the host county: Metro, Micro, or Non-Core	`Metro`
`cty_centrality`	Character	OMB role of the host county within its CBSA: Central, Outlying, or – for non-core counties	`Outlying`
`cty_msa_adj`	Character	RUCC-derived adjacency category: In Metro, Adjacent, or Non-Adjacent	`In Metro`
`rucc`	Integer	USDA Rural-Urban Continuum Code (1–9); lower values indicate larger, more urban counties	`3`

Appendix B: Additional Demonstration

Figure B1 provides two additional examples drawn from geographically and economically diverse settings—Morton, Washington and Mullins, South Carolina—to illustrate that the algorithm reliably identifies downtown boundaries across a wide range of cities. In both cases the polygon closely follows the walkable historic commercial core visible in the aerial imagery, supporting the method's generalizability.

(a) (b)

(c) (d)

Figure B1: *Downtown delineation for Morton, Washington (top row) and Mullins, South Carolina (bottom row). Panels (a) and (c) show Overture Maps points of interest within each place boundary. Panels (b) and (d) show the generated downtown polygon overlaid on satellite imagery; the polygon boundary closely follows the walkable historic commercial core in each town.*